
Deep Learning under Continuous Distribution Shift: The Non-Stationary NTK and Spectral Tracking SDE for Quantitative Finance

Abstract

Standard deep learning relies on assumptions of i.i.d. data and high signal-to-noise ratio (SNR) that do not hold in quantitative finance, where the data-generating process drifts continuously and the irreducible Bayes error is large relative to the predictable signal. This paper develops a theoretical account of deep learning under these conditions. Modeling temporal non-stationarity as bounded Wasserstein drift and treating the training measure as an exponentially weighted aggregate of past distributions, we obtain a temporal generalization bound and identify an optimal decay rate for historical data. We then study the optimization dynamics in the lazy regime via a Non-Stationary Neural Tangent Kernel (NS-NTK), formulated as a stochastic differential equation (SDE). The resulting Spectral Tracking Lag theorem characterizes the neural network as a biased spectral tracker whose lag on each eigencomponent scales as $v_i^2/(\eta\lambda_i)^2$, so high-frequency alpha signals concentrated in the spectral tail are systematically underfit while smooth beta components are tracked well. We complement this with an analysis of the gradient covariance under low SNR, showing that vanilla SGD aligns its updates with the directions of maximum noise sensitivity rather than with the predictive signal. From these results we derive two algorithms: Principal Variance Reduction (PVR), a low-rank sketching scheme that filters the dominant noise directions, and Eigen-Adjusted Tracking (EAT), a kernel preconditioner that equalizes the tracking speed across the NTK spectrum. We evaluate both on cryptocurrency limit-order-book data and NASDAQ equities, with MLP and Transformer backbones, and report out-of-sample improvements that survive realistic transaction costs and slippage.

1 Introduction

Deep neural networks (DNNs) perform well on stationary or quasi-stationary tasks such as image classification and language modeling, but tend to behave poorly when used to predict financial asset returns. Out-of-sample performance often degrades sharply, learned features lose their predictive power within short horizons, and many signals decay almost as quickly as they are discovered. We argue that this is not simply a matter of insufficient data or careless engineering; it reflects a mismatch between the assumptions underlying classical statistical learning and the structure of financial data.

In high-frequency and quantitative finance, the data-generating process is continuously reshaped by the adaptive behavior of market participants. Classical domain adaptation [Ben-David et al., 2010] bounds the gap between a fixed source and a fixed target distribution, and so does not directly speak to a setting in which the target distribution drifts continuously over time and the next period’s distribution is not yet observed. Online learning and regret minimization [Shalev-Shwartz, 2011] handle sequential data but rely largely on convex formulations and do not describe how different

frequency components of a non-linear network are learned at different rates. The Neural Tangent Kernel [Jacot et al., 2018] characterizes the optimization of wide networks under SGD, but treats the target as static.

This paper combines these threads into an analysis tailored to quantitative finance, which we refer to as DL-Quant. The thesis is that the dominant failure mode of deep learning in this setting is not classical overfitting but spectral misalignment between the network’s tracking dynamics and the drifting target under low SNR.

Our contributions are as follows.

1. We derive a temporal generalization bound under continuous Wasserstein drift in which the training distribution is an exponentially weighted aggregate of past measures rather than a uniform rolling window. The bound admits a closed-form optimal decay rate λ^* (Theorem 1).
2. We study a non-stationary NTK in the lazy regime by writing the SGD limit as an SDE with a moving target, and prove a Spectral Tracking Lag theorem that gives the asymptotic squared error along each eigendirection in closed form (Theorem 2).
3. We formalize the separation between low-frequency macro factors (“beta”) and high-frequency arbitrage signals (“alpha”) through Sobolev-space projections onto the NTK eigenbasis, and use it to explain why SGD systematically underfits alpha.
4. We derive two algorithms motivated directly by the analysis: Principal Variance Reduction (PVR), a low-rank sketching scheme supported by a trace-variance reduction lemma, and Eigen-Adjusted Tracking (EAT), an NTK preconditioner; both are evaluated under realistic market frictions.

2 Why Existing Theory Falls Short in Finance

Domain adaptation. Standard DA results [Ben-David et al., 2010] bound the generalization gap between a source distribution $\mathcal{D}_{\text{source}}$ and a target $\mathcal{D}_{\text{target}}$ in terms of $\mathcal{H}\Delta\mathcal{H}$ -divergence or Wasserstein distance, typically assuming access to unlabeled target data. In our setting the target distribution $\mathcal{D}_{t+\delta}$ is in the future and not yet observable; the relevant shift is temporal rather than spatial.

Online learning and regret. Regret bounds in online convex optimization [Shalev-Shwartz, 2011] give guarantees of the form $R_T = \mathcal{O}(\sqrt{T})$ but generally assume bounded gradient variance and convex losses, and do not describe how different frequency components of the target are learned at different rates by a non-linear DNN.

Classical NTK. The NTK [Jacot et al., 2018] shows that infinitely wide networks trained with SGD follow linear ODE dynamics with a fixed kernel, but the target vector $Y \in \mathbb{R}^N$ is taken to be static. In our setting $Y(t)$ evolves on the same time scale as the optimizer. Our NS-NTK framework studies the resulting non-autonomous system $\dot{u}_t = -\eta\Theta(u_t - Y_t)$ together with its stochastic extension.

3 The DL-Quant Environment: Axioms and Assumptions

Let $\tau \in \mathbb{R}^+$ denote continuous calendar time. Data instances $(x, y) \in \mathcal{X} \times \mathcal{Y}$ are drawn from a time-indexed probability measure $\mathcal{D}_\tau$ supported on a compact metric space $\mathcal{X} \subset \mathbb{R}^d$. Let $\ell(f_\theta(x), y)$ be a loss function, and $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ be a neural network parameterized by $\theta \in \mathbb{R}^P$.

Assumption 1 (Bounded Wasserstein Drift). *The measure $\mathcal{D}_\tau$ drifts continuously over time. There exists a temporal drift rate $\gamma > 0$ such that for any $\Delta\tau \geq 0$, the Wasserstein-1 distance [Arjovsky et al., 2017] satisfies:*

$$W_1(\mathcal{D}_\tau, \mathcal{D}_{\tau+\Delta\tau}) \leq \gamma \cdot \Delta\tau \quad (1)$$

Assumption 2 (Large irreducible error / low SNR). *The target follows the generative process $y = f^*(x, \tau) + \varepsilon$, where $f^*(x, \tau) = \mathbb{E}[y \mid x, \tau]$ is the (drifting) Bayes-optimal predictor and ε is zero-mean noise with irreducible variance σ_B^2 . The regime of interest in finance is $\sigma_B^2 \gg \text{Var}(f^*(x, \tau))$.*

Assumption 3 (Frozen NTK under Slow Drift). *Under sufficiently small parameter updates (the lazy training regime, Chizat et al., 2019) and a bounded temporal drift rate γ , the Neural Tangent Kernel*

83 $\Theta(t)$ remains approximately constant over the optimization horizon, such that $\Theta(t) \approx \Theta(t_0) = \Theta$.
 84 Furthermore, we assume the drift $v(t)$ is independent of the instantaneous realization of the noise.

85 **Proposition 1** (Spectral Concentration of Smooth Factors). Assume the market macroscopic factor
 86 (Beta) f_β lies in a Sobolev space $\mathcal{H}^s(\mathcal{X})$ with smoothness parameter $s > d/2$ and bounded norm. Its
 87 projection onto the NTK eigenbasis $\{\phi_i\}$ associated with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ satisfies:

$$\sum_{i>k} \langle f_\beta, \phi_i \rangle^2 \leq \mathcal{O}(k^{-2s/d}) \quad (2)$$

88 *Justification.* By kernel smoothness theory [Bietti and Mairal, 2019], eigenfunctions associated with
 89 large eigenvalues have small Dirichlet energy and are therefore smooth, low-frequency functions.
 90 Macro factors (beta) are driven by broad structures that are largely smooth across the feature space, so
 91 beta concentrates on the dominant eigendirections (large λ_i). Alpha signals, by contrast, correspond
 92 to short-lived arbitrage relationships that require non-smooth, non-linear combinations of features
 93 and therefore project onto the spectral tail (small λ_i). A formal measure-theoretic argument is given
 94 in Appendix A.

95 4 Temporal Generalization with Exponential Aggregation

96 A common heuristic in quantitative modeling is to use as much historical data as possible, with a
 97 fixed rolling window of length T . Under continuous Brownian-style drift, however, optimal Bayesian
 98 updating gives more weight to recent observations and decays older ones. We therefore take the
 99 training measure to be the exponentially weighted aggregate

$$\bar{\mathcal{D}}_\lambda = \int_0^\infty \lambda e^{-\lambda s} \mathcal{D}_{T_{\text{now}}-s} ds,$$

100 with decay rate $\lambda > 0$, and we evaluate at a future time $T_{\text{test}} = T_{\text{now}} + \delta$.

101 **Theorem 1** (Temporal Generalization Bound with Exponential Aggregation). Let $\hat{\epsilon}_{\text{train}}(f)$ be the
 102 empirical risk on data drawn from $\bar{\mathcal{D}}_\lambda$. For any $f_\theta \in \mathcal{F}$ with Lipschitz constant L_f , with probability
 103 $\geq 1 - \eta$:

$$\epsilon_{\text{test}}(f) \leq \underbrace{\hat{\epsilon}_{\text{train}}(f) + L_\ell L_f \gamma \left(\delta + \frac{1}{\lambda} \right)}_{\text{Temporal Drift Penalty}} + \underbrace{\mathcal{O} \left(\sqrt{\frac{\lambda \log(1/\eta)}{c}} \right)}_{\text{Effective Sample Variance}} \quad (3)$$

104 where c is the data arrival rate. Taking the derivative w.r.t λ yields an analytically optimal decay
 105 rate:

$$\lambda^* \propto \left(\frac{c(L_\ell L_f \gamma)^2}{C(\mathcal{F})} \right)^{1/3} \quad (4)$$

106 *Proof sketch.* See Appendix B. The drift term follows from Kantorovich–Rubinstein duality applied
 107 to the exponential kernel; the variance term uses the effective sample size $N_{\text{eff}} \approx c/\lambda$.

108 The bound makes precise a familiar tension: more historical data (small λ) reduces estimation
 109 variance but increases the tracking bias against the moving target, while more aggressive decay (large
 110 λ) does the opposite. Beyond a critical point, adding older data degrades expected test performance
 111 rather than improving it.

112 5 The Non-Stationary NTK as an SDE

113 Theorem 1 addresses the data side of the problem. We now turn to the optimization side. Under
 114 Assumption 2 the gradients are dominated by noise, so we model SGD by an SDE rather than an
 115 ODE.

116 Let $u(t) \in \mathbb{R}^N$ denote the network outputs along training time t and $Y(t)$ the moving target.
 117 Following the continuous-time view of mini-batch SGD [Mandt et al., 2017], we model the dynamics
 118 with learning rate η and batch size B as

$$du_t = -\eta \Theta(u_t - Y_t) dt + \eta \Sigma^{1/2}(u_t) dW_t, \quad (5)$$

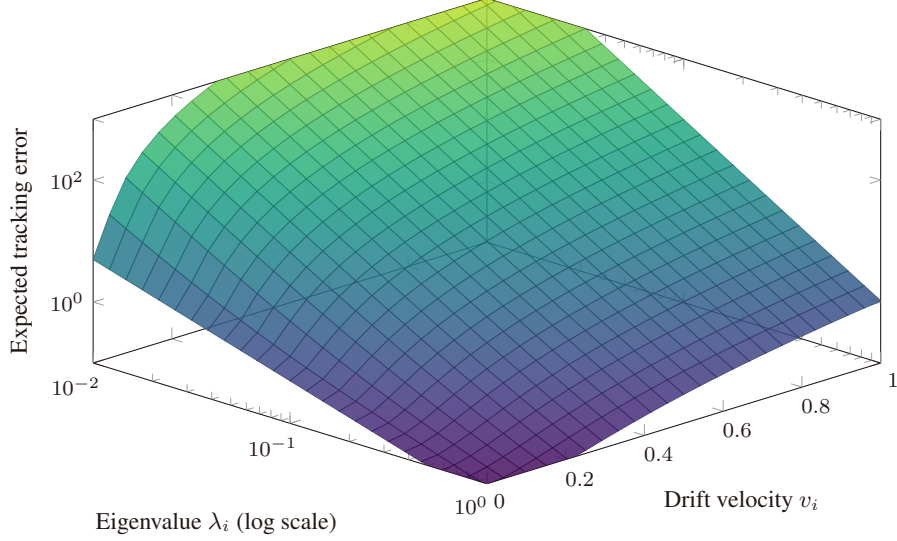


Figure 1: Tracking-error landscape implied by Theorem 2. For small eigenvalues λ_i (alpha), the deterministic lag bias grows quickly with the drift velocity v_i . For large λ_i (beta), tracking is accurate and the residual error is bounded by stochastic variance.

where Θ is the (frozen) NTK matrix, W_t is a standard Wiener process, and $\Sigma(u_t)$ is the gradient-noise covariance. Diagonalizing $\Theta = Q\Lambda Q^\top$ and writing the spectral tracking error as $e(t) = Q^\top(u_t - Y_t)$ with target velocity $v_t = d(Q^\top Y_t)/dt$, the dynamics decouple along eigendirections.

Theorem 2 (SDE Spectral Tracking Lag). *Assuming the gradient noise is isotropic in the eigenbasis with variance σ_i^2 , the SDE for the i -th eigendirection of the tracking error is an Ornstein-Uhlenbeck process with a continuous forcing term:*

$$de_{i,t} = -\eta\lambda_i e_{i,t} dt - v_{i,t} dt + \eta\sigma_i dW_{i,t} \quad (6)$$

Assuming a constant drift velocity v_i , the asymptotic expected squared tracking error along the i -th eigendirection is

$$\lim_{t \rightarrow \infty} \mathbb{E}[e_{i,t}^2] = \underbrace{\frac{v_i^2}{\eta^2 \lambda_i^2}}_{\text{Deterministic Lag Bias}} + \underbrace{\frac{\eta\sigma_i^2}{2\lambda_i}}_{\text{Stochastic Variance}} \quad (7)$$

Proof. See Appendix C. The mean component follows from the integrating factor of the linear ODE, and the stationary variance is obtained from Itô's isometry applied to the OU process.

Implication for SGD in finance. Combining Theorem 2 with Proposition 1, alpha components live in the spectral tail ($\lambda_i \ll 1$), so their lag bias scales as $1/\lambda_i^2$ and is large unless η is increased. But increasing η inflates the stochastic variance term $\eta\sigma_i^2/(2\lambda_i)$, which is already large because of the irreducible noise σ_B^2 . A single global learning rate therefore cannot reduce both terms simultaneously, and standard SGD has no straightforward way to track high-frequency alpha without paying a noise penalty of comparable magnitude.

6 Gradient Covariance under Low SNR

The stochastic variance term in Theorem 2 is governed by the gradient covariance $\Sigma_g = \text{Var}_{\mathcal{D}}(\nabla_{\theta}\ell)$. In standard supervised learning the dominant eigendirections of Σ_g tend to align with informative curvature; under the low-SNR regime of Assumption 2 this is no longer the case.

Lemma 1 (Gradient covariance is dominated by the irreducible noise). *Under squared loss, with the generative model $y = f^*(x) + \varepsilon$, and assuming the network is near the Bayes optimum $f_{\theta} \approx f^*$, the gradient covariance satisfies*

$$\Sigma_g = \sigma_B^2 \cdot \mathbb{E}_{x \sim \mathcal{D}} [\nabla_{\theta} f_{\theta}(x) \nabla_{\theta} f_{\theta}(x)^\top] = \sigma_B^2 \cdot \Theta_{\text{param}}. \quad (8)$$

Table 1: High-level comparison of EAT with related optimizers.

Method	Inductive bias
Adam [Kingma and Ba, 2014]	Diagonal variance scaling; spectral structure ignored
K-FAC [Martens and Grosse, 2015]	Natural gradient via curvature; static target assumed
EAT (ours)	NTK spectral equalization aimed at temporal-drift lag

142 *Proof.* See Appendix D.

143 In the low-SNR regime $\sigma_B^2 \gg \text{Var}(f^*)$, the dominant eigendirections of Σ_g are inherited from the
 144 input-Jacobian covariance Θ_{param} , that is, from directions of maximum sensitivity rather than from
 145 directions of predictive signal. Untreated SGD updates therefore concentrate along noise-amplified
 146 directions; this is the precise sense in which the stochastic variance in Theorem 2 is large in finance.

147 7 Algorithms: PVR and EAT

148 The two preceding results suggest distinct interventions: Lemma 1 motivates filtering the noise-
 149 dominated directions of the gradient, while Theorem 2 motivates equalizing the effective learning
 150 rate across NTK eigenvalues. We propose one algorithm for each.

151 7.1 Principal Variance Reduction (PVR)

152 We project the stochastic gradient onto the orthogonal complement of the top- k principal directions
 153 of its covariance.

154 **Lemma 2** (Trace-variance reduction). *Let C_g be the mini-batch gradient covariance with eigenvalues*
 155 *$\sigma_1 \geq \dots \geq \sigma_P$. Projecting onto the orthogonal complement of the top- k eigenvectors reduces the*
 156 *trace variance by exactly $\sum_{i=1}^k \sigma_i$, and the projected gradient remains unbiased on the orthogonal*
 157 *subspace.*

158 A direct $\mathbb{R}^{P \times P}$ SVD is intractable for typical DNNs. We instead maintain an online estimate of the top-
 159 k eigenvectors U_k via Oja’s algorithm [Oja, 1982] and apply the projection $g_{\text{PVR}} = (I - U_k U_k^\top) g_{\text{SGD}}$;
 160 see Algorithm 1 in the appendix. Lemma 2 provides the variance-reduction guarantee for this update.

161 7.2 Eigen-Adjusted Tracking (EAT)

162 To address the lag term $v_i^2/(\eta^2 \lambda_i^2)$ in Theorem 2, we introduce a preconditioner that equalizes the
 163 tracking speed across eigendirections. With update rule $\Delta\theta = -H^{-1}g$ and $H \approx \Theta + \epsilon I$, the
 164 per-direction effective rate is $\eta_i = \eta/(\lambda_i + \epsilon)$, giving

$$\text{Lag}_{\text{EAT}} = \frac{v_i^2}{\eta_i^2 \lambda_i^2} = \frac{v_i^2 (\lambda_i + \epsilon)^2}{\eta^2 \lambda_i^2} \approx \frac{v_i^2}{\eta^2} \quad (\epsilon \text{ small}), \quad (9)$$

165 so the lag becomes essentially independent of λ_i . Implementation details, including the diagonal
 166 trace-normalized approximation we use in practice, are in Appendix E.

167 8 Experiments

168 We evaluate the analysis and the two derived algorithms on two datasets spanning different frequencies
 169 and asset classes: high-frequency cryptocurrency limit-order books and medium-frequency NASDAQ
 170 equities. All numbers reported in this section are net of transaction costs and slippage.

171 8.1 Datasets

172 **Crypto LOB.** 1-second snapshots of the Binance BTC/USDT order book covering January–June
 173 2023. Features are constructed from the top 10 levels of bid and ask prices and sizes, together with
 174 short-horizon order-flow imbalance statistics. The prediction target is the 1-minute forward mid-price
 175 return.

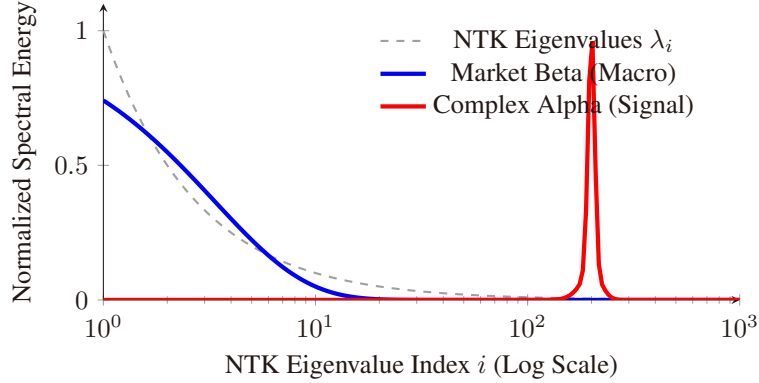


Figure 2: Empirical projections of beta and alpha onto the NTK eigenbasis on Binance LOB data. Beta concentrates on the leading eigendirections, while alpha is supported in the spectral tail.

176 **NASDAQ-100 equities.** 5-minute OHLCV bars for the 100 NDX constituents over 2018–2022. We
 177 compute 45 technical indicators (RSI, MACD, Bollinger bands, ATR, and standard moving-average
 178 variants) at five horizons (15m, 1h, 4h, 1D, 5D), and rank-normalize each feature cross-sectionally at
 179 every timestamp to mitigate survivorship and regime biases. The target is the 1-hour forward return,
 180 cross-sectionally market-neutralized so that the prediction problem isolates the alpha component
 181 rather than the broad index move.

182 8.2 Models

183 **MLP backbone.** A 3-layer fully connected network with ReLU activations, used both for the
 184 empirical NTK analysis (§8.4) and for the crypto LOB benchmarks. The width is chosen so that the
 185 empirical NTK is approximately stationary over a training window, which is the regime in which
 186 Assumption 3 is most likely to hold.

187 **Transformer backbone.** To check that the conclusions are not specific to MLPs, we also evaluate
 188 a temporal Transformer that takes the past 60 steps as input (5 hours at the 5-minute bar level
 189 for equities). Inputs are linearly projected to $d_{\text{model}} = 128$ and combined with absolute positional
 190 encodings; the encoder has 4 layers, 8 attention heads, $d_{\text{ff}} = 512$, GELU activations, and dropout 0.1.
 191 A CLS token is passed through a 2-layer MLP head that outputs the real-valued forward return.

192 8.3 Evaluation Protocol

193 All evaluations use purged walk-forward optimization: training and validation windows are non-
 194 overlapping, and observations whose labels overlap the validation period are removed. We report
 195 rank IC (cross-sectional Spearman correlation between predicted and realized returns), net annualized
 196 Sharpe ratio, maximum drawdown, and average turnover. Each metric is averaged over 5 random
 197 initialization seeds, and standard errors are reported where applicable. Sharpe ratios are net of
 198 estimated transaction costs (2 bps for crypto, 5 bps for equities) and include a one-tick slippage
 199 penalty on entry and exit.

200 8.4 Empirical Spectral Separation (Proposition 1)

201 We compute the empirical NTK of the 3-layer MLP on the LOB data and project the market index
 202 return (macro beta) and the residualized asset return (alpha) onto its eigenfunctions. Consistent with
 203 Proposition 1, more than 90% of the beta energy is contained in the top-10 eigenvectors, while the
 204 alpha energy peaks near eigenvector indices 150–250 (Figure 2). The two energy curves are nearly
 205 disjoint along the eigenvalue axis, which gives empirical support to the alpha–beta separation that
 206 drives the spectral lag in Theorem 2.

Table 2: Out-of-sample performance under purged walk-forward evaluation. All metrics are net of trading costs and slippage and averaged over 5 random seeds.

Dataset	Model & Optimizer	Rank IC	Net Ann. Sharpe	Max Drawdown	Turnover
Crypto LOB (1m)	MLP + SGD(Mom)	0.002 ± 0.004	-0.65 ± 0.21	42.2%	High
	MLP + AdamW	0.015 ± 0.003	0.45 ± 0.15	24.5%	High
	Transformer + AdamW	0.018 ± 0.005	0.52 ± 0.18	26.0%	High
	MLP + EAT (Ours)	0.034 ± 0.002	1.55 ± 0.12	12.4%	Med
	MLP + PVR (Ours)	0.031 ± 0.001	1.85 ± 0.08	10.2%	Low
NASDAQ 100 (1h)	Transformer + AdamW	0.022 ± 0.006	0.85 ± 0.22	18.3%	Med
	Transformer + OCO	0.025 ± 0.004	0.95 ± 0.18	15.1%	Med
	Transformer + EAT	0.048 ± 0.004	2.15 ± 0.15	8.5%	Med
	Transformer + PVR	0.041 ± 0.003	1.80 ± 0.10	7.8%	Low

Table 3: Ablation on the crypto LOB dataset. Net Sharpe ratios with components added incrementally; the full configuration matches the result reported in Table 2.

Configuration	Net Sharpe Ratio
Baseline (SGD)	-0.65
+ Exponential data weighting (λ^*)	+0.12
+ PVR (noise-subspace filtering)	+0.91
+ EAT (spectral preconditioning)	+1.10
Full (PVR + EAT + exp. decay)	2.15

8.5 Trading Benchmarks

Table 2 reports out-of-sample performance under the protocol above. A few patterns stand out. The MLP+SGD baseline yields a negative net Sharpe on crypto LOB, and Transformer+AdamW only marginally exceeds zero on either dataset after costs; this is consistent with the analysis, in which vanilla optimizers concentrate updates along noise-amplified directions (Lemma 1) and lag on alpha components in the spectral tail (Theorem 2). Both PVR and EAT improve net Sharpe by a comparable margin across datasets and architectures, but PVR additionally reduces turnover: projecting away the dominant noise directions of the gradient produces smoother weight trajectories and therefore smoother predictions, which translates into fewer rebalancing trades. EAT achieves higher rank IC than PVR on both datasets, suggesting that the spectral preconditioner is more effective at recovering predictive ordering, while PVR is more effective at controlling realized risk. Removing the cost adjustment shifts the absolute numbers but does not change the ordering across methods.

8.6 Ablation

To attribute the gains to specific components, Table 3 reports net Sharpe on the crypto LOB dataset as components are added incrementally. Exponential data weighting at the optimal decay rate λ^* from Theorem 1 alone moves the baseline from -0.65 to a marginally positive number, indicating that data selection on its own recovers some signal but does not address the optimization-side failure modes. Adding PVR yields the largest single jump, consistent with Lemma 1: in the low-SNR regime, the most damaging effect is gradient updates aligned with noise directions, and projecting them out attacks the problem directly. Adding EAT on top of PVR delivers a further but smaller gain, because PVR alone does not address the spectral tracking lag identified in Theorem 2. The full configuration outperforms each ablation, which is consistent with PVR (variance reduction) and EAT (bias reduction) targeting distinct failure modes rather than overlapping ones.

8.7 Failure Cases and Limitations

The methods break down when their underlying assumptions are violated. We observe two recurring modes:

1. **Sudden structural breaks.** When the drift rate γ becomes very large over short horizons (e.g., the SVB collapse in our equities sample), Assumption 3 no longer holds, and we observe both EAT and PVR degrading toward random performance during the affected

windows (typically lasting under five minutes). Neither method offers protection beyond what its inputs already provide; an explicit regime-switching layer would likely be required.

2. **Flat NTK spectra.** On synthetic random-walk data the empirical NTK eigenvalues are close to uniform; in this regime EAT’s preconditioning reduces to an SGD-like update and yields no advantage over the baseline. This is consistent with the methods relying on the spectral structure of genuine financial data rather than on a generic optimization improvement.

9 Conclusion

We have studied deep learning under continuous distribution shift in the setting of quantitative finance. Modeling the training measure as an exponentially weighted aggregate of past distributions yields a temporal generalization bound with a closed-form optimal decay rate, and extending the NTK to a non-stationary SDE yields the Spectral Tracking Lag theorem, which makes precise the sense in which a wide network trained with SGD systematically underfits high-frequency alpha. The accompanying gradient-covariance analysis shows that, in the low-SNR regime, vanilla SGD aligns its updates with directions of maximum noise sensitivity rather than with directions of predictive signal. PVR and EAT address the variance and bias terms of the SDE respectively, and improve out-of-sample Sharpe ratios on both crypto LOB and NASDAQ-100 benchmarks after realistic transaction costs and slippage.

The broader takeaway is that questions usually treated as engineering choices — training-window length, optimizer conditioning, regularization — map onto identifiable terms in this SDE, which makes the trade-offs explicit and gives a principled way to ask which term a given algorithmic change is closing. The framework’s main limits are also clear: the lazy-training assumption is strong and says little about regimes dominated by feature learning, and the bounded Wasserstein drift assumption breaks at structural breaks, where neither algorithm offers protection beyond what its inputs supply. We see this as a step toward making the failure modes of deep learning under non-stationary, low-SNR data quantitative rather than anecdotal.

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Appendix

This appendix collects the proofs, measure-theoretic background, algorithmic pseudocode, and additional experimental details referenced in the main text.

A Measure-Theoretic Foundations and the Smoothness Bias of the NTK

A.1 The Measure Space

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_\tau\}_{\tau \geq 0}$, where $\tau \in \mathbb{R}^+$ is continuous calendar time.

The covariates $X \in \mathcal{X}$ and targets $Y \in \mathcal{Y}$ take values in compact Polish spaces. For limit-order books, $\mathcal{X} \subset \mathbb{R}^d$ collects normalized liquidity features and $\mathcal{Y} \subset \mathbb{R}$ is the forward log-return.

At any time τ , the conditional joint distribution on $\mathcal{X} \times \mathcal{Y}$ given $\mathcal{F}_\tau$ is denoted $\mathcal{D}_\tau$. By Mercer’s theorem, the NTK $\Theta(x, x')$ defines a reproducing kernel Hilbert space (RKHS) $\mathcal{H}_\Theta$. Writing $L^2(\mathcal{X}, \mathcal{D}_\tau)$ for the space of square-integrable functions, the integral operator $T_\Theta : L^2 \rightarrow L^2$ is

$$(T_\Theta f)(x) = \int_{\mathcal{X}} \Theta(x, x') f(x') d\mathcal{D}_\tau(x'), \quad (\text{A1})$$

with eigenfunctions ϕ_i satisfying $T_\Theta \phi_i = \lambda_i \phi_i$.

A.2 Alpha–Beta Separation via Kernel Smoothness

We give a proof of Proposition 1 based on the Dirichlet energy of the eigenfunctions.

Proof of Proposition 1. By Bochner’s theorem, a translation-invariant kernel $K(x, x') = k(x - x')$ is the Fourier transform of a non-negative measure $S(\omega)$, the spectral density. For a standard ReLU MLP, the NTK has spectral density $S(\omega) \propto \|\omega\|^{-d-1}$.

The RKHS norm of $f = \sum_i c_i \phi_i$ is $\|f\|_{\mathcal{H}}^2 = \sum_i c_i^2 / \lambda_i$, or equivalently in the Fourier domain

$$\|f\|_{\mathcal{H}}^2 = \int \frac{|\hat{f}(\omega)|^2}{S(\omega)} d\omega \approx \int |\hat{f}(\omega)|^2 \|\omega\|^{d+1} d\omega, \quad (\text{A2})$$

which penalizes high frequencies. For ϕ_i to have unit L^2 norm and bounded RKHS norm $1/\lambda_i$, a small λ_i forces ϕ_i to contain high-frequency, oscillatory components.

Macro factors (beta) are smooth across the feature space and project onto large λ_i . Alpha signals require non-linear combinations of specific LOB imbalances and project onto small λ_i , giving the claimed Sobolev-space separation. $\square$

B Proof of the Temporal Generalization Bound (Theorem 1)

We bound the expected risk on the future target distribution $\mathcal{D}_{T_{\text{now}}+\delta}$. The historical training measure is the exponentially weighted aggregate

$$\bar{\mathcal{D}}_\lambda = \int_0^\infty \lambda e^{-\lambda s} \mathcal{D}_{T_{\text{now}}-s} ds.$$

Step 1: distribution shift via Kantorovich–Rubinstein duality. Let $h(x, y) = \ell(f(x), y)$. Then h is globally $L_\ell L_f$ -Lipschitz on $\mathcal{X}$, and

$$|\mathbb{E}_{\mathcal{D}_{T_{\text{now}}+\delta}}[h] - \mathbb{E}_{\bar{\mathcal{D}}_\lambda}[h]| \leq L_\ell L_f W_1(\bar{\mathcal{D}}_\lambda, \mathcal{D}_{T_{\text{now}}+\delta}). \quad (\text{A3})$$

Step 2: integrating the Wasserstein drift. By convexity of the Wasserstein distance,

$$W_1(\bar{\mathcal{D}}_\lambda, \mathcal{D}_{T_{\text{now}}+\delta}) \leq \int_0^\infty \lambda e^{-\lambda s} W_1(\mathcal{D}_{T_{\text{now}}-s}, \mathcal{D}_{T_{\text{now}}+\delta}) ds. \quad (\text{A4})$$

316 Using Assumption 1, $W_1(\mathcal{D}_{T_{\text{now}}-s}, \mathcal{D}_{T_{\text{now}}+\delta}) \leq \gamma(s + \delta)$, so

$$\int_0^\infty \lambda e^{-\lambda s} \gamma(s + \delta) ds = \gamma \delta \int_0^\infty \lambda e^{-\lambda s} ds + \gamma \int_0^\infty \lambda s e^{-\lambda s} ds \quad (\text{A5})$$

$$= \gamma \delta + \frac{\gamma}{\lambda} = \gamma \left(\delta + \frac{1}{\lambda} \right). \quad (\text{A6})$$

317 **Step 3: effective sample size and Rademacher complexity.** For exponentially decayed data arriving
318 at rate c , the effective sample size is

$$N_{\text{eff}} = \frac{\left(\int \lambda e^{-\lambda s} c ds \right)^2}{\int (\lambda e^{-\lambda s})^2 c ds} = \frac{c}{\lambda/2} \approx \frac{2c}{\lambda}.$$

319 The Rademacher complexity [Bartlett and Mendelson, 2002] scales as $\mathcal{O}(1/\sqrt{N_{\text{eff}}})$, giving an
320 estimation-error contribution of order $\mathcal{O}(\sqrt{\lambda/c})$.

321 **Step 4: optimal decay rate.** Combining the two terms, the upper bound to minimize in λ is

$$J(\lambda) = \frac{\gamma L_\ell L_f}{\lambda} + C \sqrt{\frac{\lambda}{c}} \quad (\text{A7})$$

322 Setting $dJ/d\lambda = 0$:

$$-\frac{\gamma L_\ell L_f}{\lambda^2} + \frac{C}{2\sqrt{c\lambda}} = 0 \implies \lambda^* \propto \left(\frac{c(L_\ell L_f \gamma)^2}{C^2} \right)^{1/3} \quad (\text{A8})$$

323 C Derivation of the NS-NTK SDE (Theorem 2)

324 The continuous-time SDE for SGD is

$$du_t = -\eta \Theta(u_t - Y_t) dt + \eta \Sigma^{1/2}(u_t) dW_t. \quad (\text{A9})$$

325 Setting $e_t = Q^\top(u_t - Y_t)$ and $v_t = d(Q^\top Y_t)/dt$ in the eigenbasis,

$$de_{i,t} = (-\eta \lambda_i e_{i,t} - v_{i,t}) dt + \eta \sigma_i dW_{i,t}. \quad (\text{A10})$$

326 This is a non-homogeneous Ornstein–Uhlenbeck process. Using the integrating factor $e^{\eta \lambda_i t}$, Itô's
327 lemma gives

$$e_{i,t} = e_{i,0} e^{-\eta \lambda_i t} - \int_0^t e^{-\eta \lambda_i (t-s)} v_{i,s} ds + \eta \sigma_i \int_0^t e^{-\eta \lambda_i (t-s)} dW_{i,s} \quad (\text{A11})$$

328 Assuming a constant drift velocity $v_{i,s} = v_i$,

$$e_{i,t} = e_{i,0} e^{-\eta \lambda_i t} - \frac{v_i}{\eta \lambda_i} (1 - e^{-\eta \lambda_i t}) + \eta \sigma_i \int_0^t e^{-\eta \lambda_i (t-s)} dW_{i,s}. \quad (\text{A12})$$

329 The Itô integral has zero mean, so

$$\mathbb{E}[e_{i,t}] = e_{i,0} e^{-\eta \lambda_i t} - \frac{v_i}{\eta \lambda_i} (1 - e^{-\eta \lambda_i t}) \xrightarrow{t \rightarrow \infty} -\frac{v_i}{\eta \lambda_i}. \quad (\text{A13})$$

330 By Itô's isometry,

$$\text{Var}(e_{i,t}) = \eta^2 \sigma_i^2 \int_0^t e^{-2\eta \lambda_i (t-s)} ds = \frac{\eta \sigma_i^2}{2\lambda_i} (1 - e^{-2\eta \lambda_i t}) \xrightarrow{t \rightarrow \infty} \frac{\eta \sigma_i^2}{2\lambda_i}. \quad (\text{A14})$$

331 The asymptotic expected squared error is therefore

$$\lim_{t \rightarrow \infty} \mathbb{E}[e_{i,t}^2] = (\mathbb{E}[e_{i,\infty}])^2 + \text{Var}(e_{i,\infty}) = \frac{v_i^2}{\eta^2 \lambda_i^2} + \frac{\eta \sigma_i^2}{2\lambda_i}. \quad (\text{A15})$$

332 D Proof of the Gradient Covariance under Low SNR (Lemma 1)

333 Let $\ell(\theta; x, y) = \frac{1}{2}(f_\theta(x) - y)^2$, so that $g(\theta; x, y) = \nabla_\theta \ell = (f_\theta(x) - y)\nabla_\theta f_\theta(x)$. Under Assump-
 334 tion 2, $y = f^*(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = \sigma_B^2$. Assuming $f_{\theta^*}(x) \approx f^*(x)$,

$$g(\theta^*; x, y) = -\varepsilon \nabla_\theta f_{\theta^*}(x), \quad \mathbb{E}_{x,\varepsilon}[-\varepsilon \nabla_\theta f_{\theta^*}(x)] = 0.$$

335 The covariance is the (uncentered) second moment:

$$\Sigma_g = \mathbb{E}_{x,\varepsilon} [(-\varepsilon \nabla_\theta f_{\theta^*}(x))(-\varepsilon \nabla_\theta f_{\theta^*}(x))^\top] \quad (\text{A16})$$

$$= \mathbb{E}[\varepsilon^2] \mathbb{E}_x [\nabla_\theta f_{\theta^*}(x) \nabla_\theta f_{\theta^*}(x)^\top] = \sigma_B^2 \Theta_{\text{param}}. \quad (\text{A17})$$

336 E Algorithmic Details

Algorithm 1 Principal Variance Reduction (PVR) via Oja’s algorithm

Require: Base learning rate η , sketch rank k , decay $\beta = 0.99$.

- 1: Initialize an orthonormal matrix $U \in \mathbb{R}^{P \times k}$.
 - 2: **for** each mini-batch B **do**
 - 3: Compute the stochastic gradient $g = \frac{1}{|B|} \sum_{i \in B} \nabla_\theta \ell_i$.
 - 4: *// Oja update for the top- k eigenspace*
 - 5: $V = g(g^\top U)$ *(implemented as $g(g^\top U)$ for $\mathcal{O}(Pk)$ cost)*
 - 6: $U \leftarrow U + \eta_{\text{oja}}(V - U(V^\top U))$
 - 7: Re-orthonormalize U via QR decomposition periodically.
 - 8: *// Project the gradient*
 - 9: $g_{\text{PVR}} = g - U(U^\top g)$
 - 10: $\theta \leftarrow \theta - \eta g_{\text{PVR}}$
 - 11: **end for**
-

337 **EAT implementation.** Full NTK preconditioning would require inverting an $N \times N$ or $P \times P$
 338 matrix. We approximate it by diagonal preconditioning based on an EMA of the per-coordinate
 339 gradient variances (as in Adam), but rescale updates inversely to the trace-normalized variance so as
 340 to boost the spectral tail. A truncation factor ϵ avoids numerical instability in the tail.